

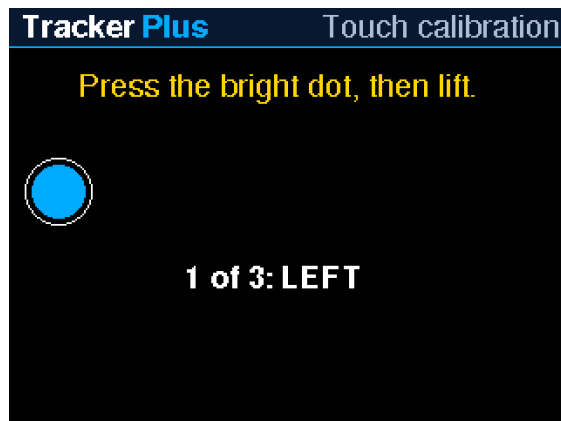
Tracker Plus

How-to guide — Sports, Soccer, Stock Ticker and Aircraft Radar on one CYD2usb (ESP32-2432S028C)

1. Flash it

- Open **TrackerPlus_9.ino** in the Arduino IDE. The four tracker tabs (tp_sports, tp_soccer, tp_stocks, tp_radar) open with it.
- Libraries (Library Manager): **GFX Library for Arduino**, **ArduinoJson 7**, **TJpg_Decoder** (Bodmer).
- Board: **ESP32 Dev Module**. Tools > Partition Scheme: **Minimal SPIFFS (1.9MB APP with OTA/190KB SPIFFS)**. The real layout comes from **partitions.csv** in the sketch folder (keep it there): two program slots for Wi-Fi updates plus 512 KB for Soccer logos. Tools > Erase All Flash Before Sketch Upload: **Disabled**.
- Upload. Serial Monitor at 115200 shows *TRACKER PLUS BUILD 9*.

2. First start



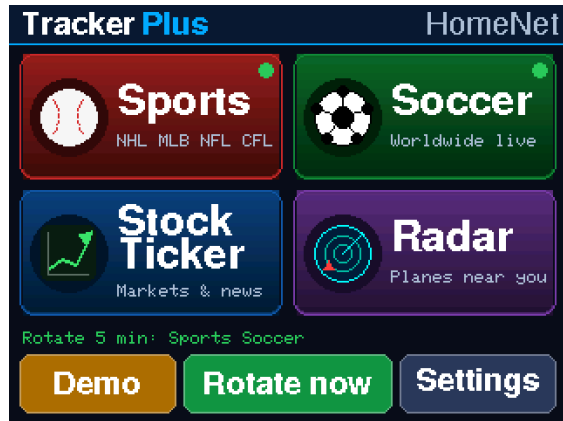
Touch calibration: press each dot, then lift



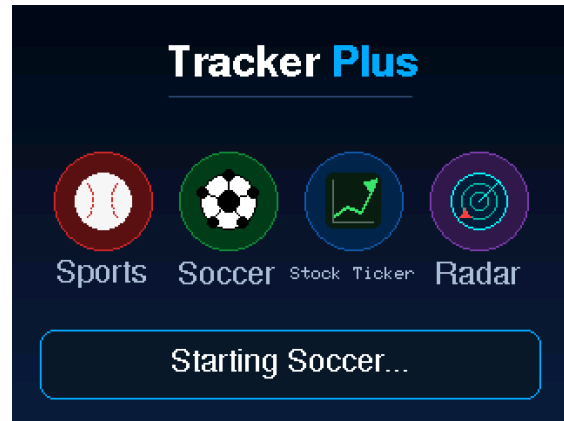
Wi-Fi setup

- A fresh unit calibrates the touch screen first, then shows the Wi-Fi screen. Type your network or tap **SCAN**, then **OK**.
- Calibration and Wi-Fi are saved once and shared by all four trackers.

3. Home menu



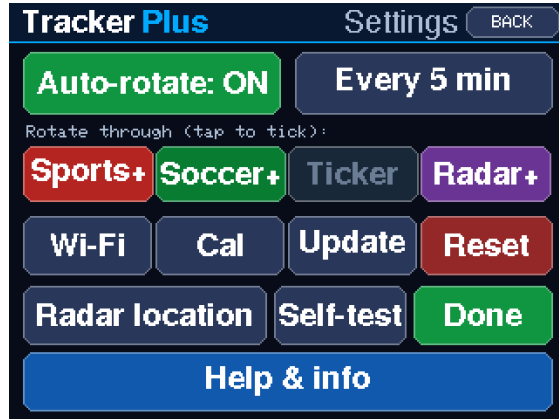
Home: tap a tile to open that tracker



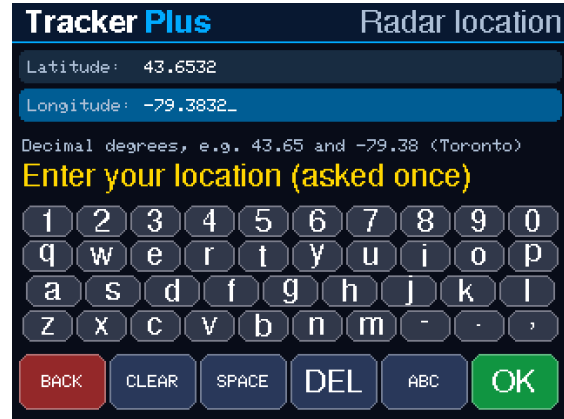
Start-up / switching screen

- Tap **Sports**, **Soccer**, **Stock Ticker** or **Radar**. Only one tracker runs at a time.
- Switching restarts the board quietly (about 3–5 s) so the tracker you open always gets all the memory.
- **Please be patient when opening a tracker.** Each time you open one (or go back to Home), it reconnects and downloads fresh data, so it can take several seconds up to about a minute before everything is filled in. Sports and Soccer take longest the first time, while they load schedules and team logos.
- To come back: open the tracker's own menu and tap **Home**. It replaces the old *Touch cal* button there, since calibration is now under Tracker Plus Settings. Radar has a **HOME** button at the top left.
- **Stock Ticker:** clock, weather, stocks, crypto and news. Nothing to enter: location and time zone are found automatically. Tap the top right for its Settings, where **Home** is.
- **Radar:** the first time, it asks for your latitude and longitude (e.g. 43.65 and -79.38). It is saved, and you can change it in Settings > *Radar location*.
- Your teams, map view, time zone and so on are kept inside each tracker, as before.

4. Settings and auto-rotate



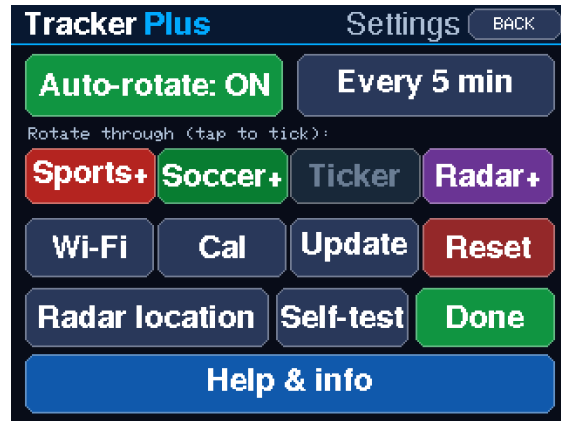
Settings



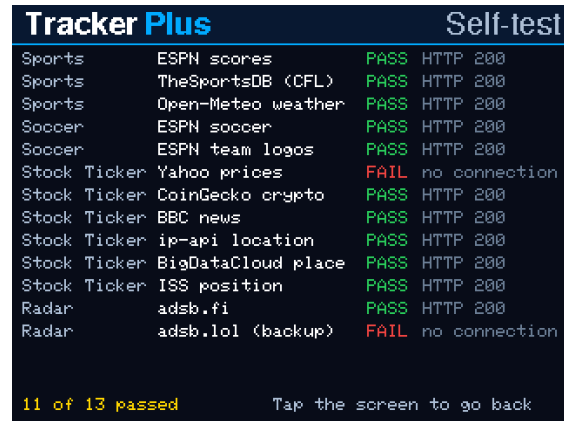
Radar location (asked once)

- **Auto-rotate ON/OFF:** when on, the board moves to the next ticked tracker after the chosen time.
- **Every N min:** tap to cycle through 2, 5, 10, 15 and 30 minutes.
- **Rotate through:** tap a tracker to add it (it shows a +, e.g. *Soccer+*) or take it out (dim, no +). Stock Ticker is shown as *Ticker* there. A green dot on the Home tile shows it is in the rotation.
- Rotation waits while you are in a menu, while a goal or score flash is showing, and while one of your Sports or Soccer teams is playing live.
- Each switch in the rotation reloads that tracker's data, so every tracker needs a few moments (sometimes up to a minute) to fill in after it appears. This is normal. Choose a longer time (10 min or more) to spend less time waiting and more time watching.
- With auto-rotate on, the board starts rotating at power-up. **Home** pauses it, and **Rotate now** on the Home screen restarts it.
- **Wi-Fi**, **Cal** (touch calibration) and **Radar location** can be changed here at any time.

5. Self-test (run it on every unit before it ships)



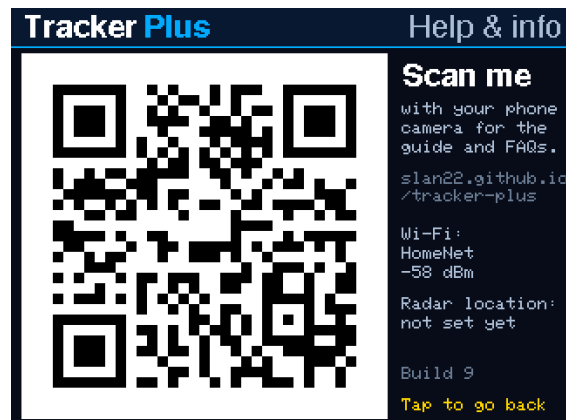
Settings: Self-test button



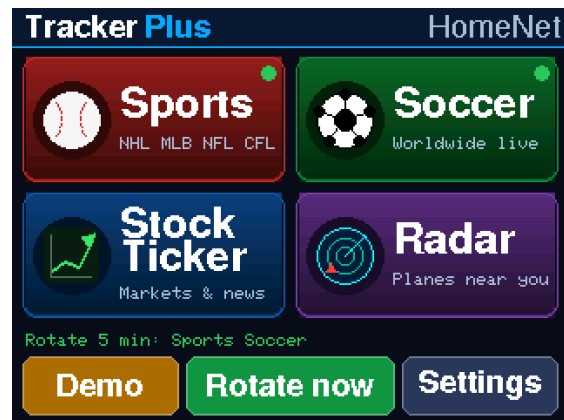
Self-test results

- Settings > **Self-test** checks every internet source the four trackers use, one by one (about a minute).
- **PASS** means that source answered. **FAIL** shows the reason (an HTTP code, or *no connection*). Tap the screen to go back.
- Radar has two sources (adsb.fi, with adsb.lol as backup), so one FAIL there still leaves Radar working. Stock Ticker gets its prices from Yahoo.
- The results are also printed in the Serial Monitor (115200).

Help & info



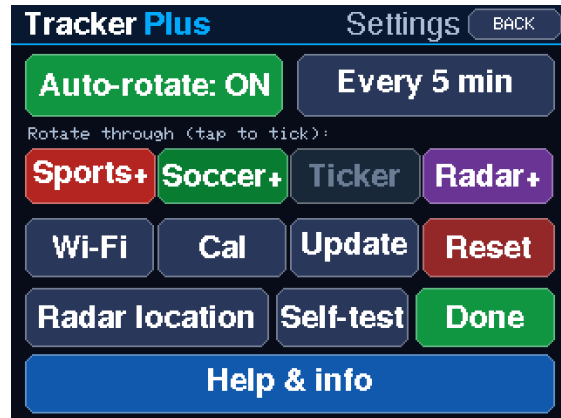
Settings > Help & info



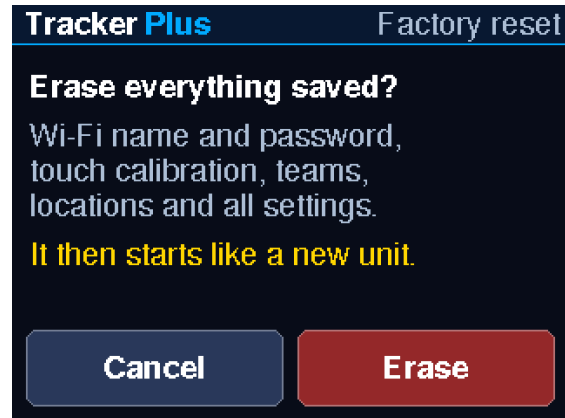
Home

- Settings > **Help & info** shows a QR code. Scan it with a phone camera to open this guide and the FAQs online at slan22.github.io/tracker-plus - nothing to print.
- The same screen shows the Wi-Fi name and signal, the Radar location and the build number. Tap to go back.

Before a unit ships: factory reset



Settings: Reset (red)



Factory reset: Cancel or Erase

- Test the unit on your Wi-Fi (Self-test, open each tracker), then Settings > **Reset** > **Erase**.
- It erases the Wi-Fi name and password, touch calibration, teams, Radar and Stock Ticker locations and every setting, then restarts like a new unit (calibration, then Wi-Fi setup).
- After the reset, switch it off and box it. Do not connect it to your Wi-Fi again.
- No Wi-Fi name, password or personal detail is in the code files; they only live on a board after someone types them in.

6. Software updates over Wi-Fi

- Settings > **Update** checks for a newer build. Nothing is installed unless the owner taps **Install**. If a download fails, the board keeps running the build it had.
- Updates come from the latest release of github.com/slan22/tracker-plus. The repository must be **public** and must never be deleted or renamed, because every board has this address built in.
- Only boards flashed with build 5 or later can update over Wi-Fi. Earlier boards need one USB flash first. Updates keep the owner's Wi-Fi and settings.

How to publish an update

- **Once:** on GitHub create a public repository named **tracker-plus**.
- **1.** In the .ino, raise **TP_BUILD_NUM** by one (e.g. 5 to 6) and change TP_BUILD to match.
- **2.** Arduino IDE: Sketch > **Export Compiled Binary**. In the sketch folder open build > esp32.esp32.esp32 and copy the file ending **.ino.bin** (not merged, not bootloader). Rename the copy to exactly **TrackerPlus.bin**.
- **3.** Make a text file named exactly **version.txt** containing only the new number (e.g. 6).
- **4.** GitHub > tracker-plus > **Releases** > **Draft a new release**. Tag: v6. Attach **TrackerPlus.bin** and **version.txt**. Click **Publish release**.
- **5.** On a board: Settings > Update > Install. Test on your own board before telling buyers.
- The program must stay under 1.69 MB to fit a program slot (build 9 is 1.55 MB).
- When the board starts, it tries your Wi-Fi up to 3 times (about 30 s) before showing the Wi-Fi screen.
- If the Wi-Fi screen is left alone for a minute (for example after a power cut, while the router is still starting), it tries the saved network again by itself, every minute.
- Radar stays on its screen when Wi-Fi drops and carries on when it is back.
- Radar near a busy airport: every plane is drawn, but only the 6 closest get a name and height (fewer if they would overlap). **NEXT** steps through the planes from closest to farthest; the chosen one is ringed in yellow with its label.

8. Demo

- Tap **Demo** on Home. It shows each tracker with sample data: Sports (about 45 s), Soccer (45 s), Stock Ticker with live data (45 s), then Radar with sample aircraft (45 s). Then it returns to Home.
- Tap **Home** in any tracker's menu to stop the demo early. Wi-Fi is needed because the demo uses real logos and live Stock Ticker data.

9. Troubleshooting

- Taps landing in the wrong place: Settings > **Cal** (touch calibration). Press the centre of each dot and hold it still for a moment.
- Wi-Fi fails: after 3 tries the Wi-Fi screen comes back with the reason shown, and keeps retrying the saved network every minute.
- A tracker looks empty or says *Loading* right after opening: give it a minute, it is downloading fresh data. If it still shows no data after that, run Settings > **Self-test** to see which source is not answering.

- Sports gets NHL, MLB and NFL games from ESPN (every game, preseason included). CFL comes from TheSportsDB.
- Program too big when compiling: set Partition Scheme to **Minimal SPIFFS (1.9MB APP with OTA)** and keep **partitions.csv** in the sketch folder.
- First start after flashing build 5 or later: the Serial Monitor may show one *LittleFS mount failed* error when Soccer opens. Soccer formats its logo storage and carries on; it only happens once.
- After a crash or power cut, the board comes back to Home (or starts rotating if auto-rotate is on).